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Project Title:	Operational Performance Effects and Neurophysiology in Partial Gravity (OPEN-PG)		
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Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	Yes
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Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
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Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Ivkovic, Vladimir Ph.D. (Massachusetts General Hospital) Zhang, Quan Ph.D. (Massachusetts General Hospital)		
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Task Description:

Background: Understanding human performance under various partial-gravity loadings is critical to NASA's crewed mission strategies. For example, current vehicle and mission designs for Mars landings require the crew to use robotic teleoperation within 24 hours of landing—unaided from Earth—to connect landing craft power systems with pre-positioned power generators. This design requires the crew to perform complex sensorimotor operations to maintain life support, as soon as possible after a gravitational transition, and in an unfamiliar partial-gravity setting. In addition, missions to the Moon and Mars will require considerable activity on the planetary surfaces in reduced gravity, and little is known about postural stability and likelihood of falls in such settings. The neurophysiological changes associated with such activities is similarly poorly understood but may provide insights into workload or even provide predictive value for performance in hypogravity settings. Addressing these gaps will in part require integrated assessment of operational and sensorimotor performance alongside neurovestibular and neurophysiological responses during exposure to various gravitational loads.
Aim 1: Characterize and quantify changes in operationally-relevant sensorimotor and vestibular performance as a function of gravitational load.

Aim 2: Characterize and quantify changes in physiology—particularly in brain function and autonomic activation during behavioral performance—as a function of gravitational load.

Aim 3: Develop a model to predict behavioral performance and neurophysiological responses under different gravitational loads based on preflight ground testing data.

Hypotheses: (Hyp1) We predict a monotonic but non-linear relationship between Robotics On-Board Trainer-r (ROBoT-r) performance and gravitational load, with larger departures from 1g leading to more impaired performance. (Hyp2) Behavioral alterations will be paralleled by physiological changes at different gravity loads, including activation of prefrontal and vestibular cortex, and autonomic nervous system activation. (Hyp3) Ground-based challenges to the vestibular system will induce detectable neurophysiological responses, and the amplitude of these responses (i.e., an indicator of individual "sensitivity" to these provocations) will help (3a) predict neurophysiological responses in-flight, and (3b) predict behavioral performance in flight.

Deliverables: Overall, our project sought to characterize (1) operationally-relevant performance and (2) neurophysiological responses as a function of gravity load, as well as (3) providing models to predict performance and neurophysiological impacts of partial gravity based on preflight-data. This work has the potential to identify individuals who are particularly resilient to altered gravity, and will be key for planning future exploration-class missions where survival will depend on the operational capabilities of astronauts in partial-gravity environments.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

These parabolic flights will help fill critical knowledge gaps regarding human exposure to fractional-gravity conditions. Specifically, our project will address gaps regarding operational performance, neurophysiological status, individual sensitivity to different gravitational loads between 0-1g, as well as prediction of behavioral performance and physiological responses to partial gravity. In addition to filling key gaps surrounding human performance of operationally-relevant tasks in partial gravity, this work may provide a method to help identify crewmembers who are particularly resilient for performing particular tasks under novel gravity loadings. The results have the further benefit of providing a better understanding of the role of disorientation in Earth-based operational performance. This is relevant not only to fighter pilots, but to task performance by individuals with neurological or medical conditions that adversely affect the vestibular system (e.g., stroke, inner ear infections).

We completed all data collection as planned, with a 98.3% data collection rate for all measurements (teleoperation, postural balance, multi-modal neurophysiology, and survey data). The major takeaways from this OPEN-PG project are as follows:
General: • Novel Data Acquisition: Our data collection efforts were the first to simultaneously measure multiple hemoglobin species (oxy-, deoxy-, and total-Hb)—which are closely related to cerebral blood volume (CBV)—synchronized with changes in cerebral water content (H2O)—which is closely related to cerebrospinal fluid (CSF) volume. This will provide novel insights into cerebral fluid dynamics across multiple gravity conditions.

Aim 1: Operationally-relevant characterization • Postural Balance Performance: We thoroughly quantified changes in static postural balance as a function of gravitational load and ground-based neurovestibular provocations. Multiple balance-related outcome measures were significantly and substantially affected by g-load. Several changes in outcomes appeared to accelerate in magnitude towards the lowest partial-gravity level (0.25g). Since no parabolas were conducted with Moon or Mars gravity levels (0.16g and 0.38g, respectively), we can only estimate the operationally-relevant balance performance in those settings. For Mars gravity, this is an interpolation process and perhaps a reasonable estimate can be made. For the Moon, however, we had no force plate measurement at 0g (of course), and the next measure was at 0.25g, making estimation of balance in Lunar gravity an extrapolation on an accelerating curve. Any such estimate would be highly suspect. Given the near-term prospect of operations on the Moon, direct measurements during Lunar-gravity parabolas are highly recommended to better understand balance in that setting. • ROBoT-r Sensorimotor Performance: ROBoT-r performance was largely unaltered in hypogravity, except in the 0g setting. The 0g setting does add the challenge of remaining in place (i.e., not floating away) while performing ROBoT-r operations, and, although a harness prevented participants from floating away entirely, there was undoubtedly a distraction or added challenge with maintaining one's position and executing the controller movements in 0g. The testing periods were brief (<20 sec), so it is still not known how performance might be affected in hypogravity over longer periods. However, over these short periods, and going the more intense "final approach" of the ROBoT-r track-and-capture task, operational performance was relatively good. What did appear to be affected, however, was learning—little to no learning was indicated across all sessions. This highlights the importance of thorough and up-to-date pre-training on any tasks that will be conducted during intense periods.

Aim 2: Neurophysiology characterization • HR: Heart rate decreased at all partial-g levels, although it remained steady at 0g compared to 1g. This is generally consistent with previous work. • HRV: Heart rate variability was largely unchanged in hypogravity, which was unexpected. We hypothesize this arose because volunteers were highly experienced flyers, including multiple fighter pilots who would be less expected to exhibit large sympathetic responses to the mild flight maneuvers generated in this study. • Cerebrovascular Pulsatility: Cerebrovascular pulsatility showed intriguing responses that were largely consistent across hemispheres and hypogravity conditions. Pulsatility increased in

Task Progress:	0g, which matches our prior hypotheses. Interestingly, our ground-based provocations reduced pulsatility. If true, this could be a useful feature, if indeed increased pulsatility is a potential contributor to spaceflight associated neuro-ocular syndrome (SANS) as previously hypothesized. • Cerebral Blood Flow: There was very strong evidence of substantially increased blood flow to the antero-lateral prefrontal cortex associated with hypogravity, and that this increase became substantially larger at the lowest gravity loads (0.25g). This finding may also have relevance to SANS, but measurements over much longer periods in hypogravity would be necessary to determine its importance. • Cerebral Blood Volume: There was similarly strong evidence of increased blood volume in the brain associated with hypogravity conditions, with larger blood volumes at lower gravity loads and a tendency towards a nonlinearly bigger effect at 0g. This also has relevance to SANS, given the venous congestion hypothesis for that condition. Aim 3: Prediction • To date, prediction of operational performance from our ground data has been largely unsuccessful. This replicates numerous prior studies attempting similar predictions. Additional work on this dataset is certainly possible, but was beyond the scope of this project. Recommendations: Finally, we provide a few recommendations to be considered for future work. • Direct Moon/Mars Characterization: While the distribution of g-levels (0, ¼, ½, ¾, 1g) “spanned” the hypogravity range, our findings suggest there will be challenges generalizing those to the current gravity loads of interest, including Martian (0.38g) and particularly Lunar (0.16g) gravity levels. It is recommended that parabolic flight investigations be conducted at these two gravitational loads, specifically, to identify any unexpected features at either load. We recommend testing at the Lunar g-load in particular, for the following two reasons: o The Moon is our next stopping point for humans, and knowing as much as we can before we arrive will help reduce health and performance risks during Lunar missions. o Estimating performance from existing data is largely an extrapolation process on an accelerating curve. This is a particularly fraught context in which to obtain accurate estimates. • Prediction: Given the importance of being able to predict who will have their operationally-relevant performance more or less affected by hypogravity exposure, we recommend further investigation of this topic. Options include: o Combine datasets across studies to enable prediction based on a broader range of predictor variables. o Applying machine learning approaches that can not only seamlessly combine predictions from a wide array of predictor variables, but also implicitly account for interactions among such variables, which are almost certainly important. • Analog Development: We deployed 3 provocations (GVS, head pitch, and vection stimuli) that were intended as “analogs” of the vestibular disruption associated with hypogravity during space missions. In essentially no case did the balance performance or physiology measures following these disruptions appear similar to the hypogravity cases. GVS came closest, but still had a modest effect relative to actual hypogravity. Having characterized balance and (neuro)physiological responses to hypogravity in this study, one could use such information to help guide the development of new analogs (e.g., GVS with stronger stimuli, or more robust vection/head tilt procures, or whole-body spinning or other disruptions) that more realistically reflect the disruptions found in hypogravity. • Individual Variability: Teleoperation, balance, (neuro)physiology, and responses to provocations are all often highly variable across individuals. It was beyond the scope of this project to do a detailed evaluation of individual variability, but given that many spaceflight phenomena affect some people more than others, a detailed assessment of individual variability is warranted. It is possible that such an investigation could help explain some of the unexpected features in our data. The OPEN-PG work provided an excellent starting point for investigating and thoroughly characterizing operationally-relevant performance in partial gravity settings. Balance was quite adversely affected; whereas, teleoperation performance was minimally affected. Characterizing the details of the available gravity loads was a solid first start, and provides an initial roadmap for how to investigate Lunar and Martian gravity loads and what types of tasks and measures should be considered in particular.
Bibliography Type:	Description: (Last Updated: 07/08/2026)
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